Jesse Silva

jesse_silva@student.uml.edu | Peabody, MA | (978)-390-1768 | linkedin.com/in/jesse-silva1204/

OBJECTIVE

Computer Engineering student with semiconductor manufacturing experience and hands-on embedded hardware, PCB design, and firmware development seeking an embedded systems or hardware engineering position.

ENGINEERING EXPERIENCE

Applied Materials (Formerly ASMPT Nexx) **Billerica, MA**
Quality Engineering Co-Op July, 2026 - Present

- Performed dimensional, flatness, and surface inspections of precision semiconductor-equipment components using engineering drawings and GD&T; documented nonconformances to support disposition decisions.
- Utilized UniPoint and Made2Manage ERP system for managing nonconformances and inventory transactions.
- Conducted measurement-system R&R testing across **16 measurement zones and 9 pressure conditions (10–40 PSI)**; analyzed results using Excel and Power Query.
- Performed final shipment validation on **2 semiconductor manufacturing machines**, verifying compliance with customer specifications and quality requirements.

TECHNICAL SKILLS

- **Programming:** C/C++, VHDL, ARM Assembly, RISC-V Assembly
- **EDA / Design:** Altium Designer, KiCad, Quartus, MATLAB, NI Multisim
- **Embedded:** STM32, PIC18, ESP32, I²C, SPI, UART, SWD
- **Test & Measurement:** Oscilloscopes, DMMs, Function Generators, Logic Analyzers, JTAG/SWD Debuggers
- **Interpersonal:** Technical Communication, Cross-Functional Collaboration, Team Leadership, Project Coordination

PROJECTS

SMTA Bright Electronics Manufacturing Competition Robot | 2026

- Designed a custom PIC18-based line-following robot PCB integrating **7 reflectance sensors, 2 wheel encoders, an IMU, and dual-motor control**.
- Developed hardware for sensor acquisition, motion control, and autonomous line tracking and recovery as part of a multidisciplinary student team.
- **Tied for 1st place with Michigan Technological University in PCB design**, advancing into the assembly and competition rounds.

Custom Embedded PCB Business Cards — Versions 1 & 2 | 2026

- Designed an STM32G030-based PCB integrating a **6-axis IMU and multiplexed 8×8 LED matrix**; developed embedded C firmware for motion-reactive animations and gameplay, text rendering, and low-power operation.
- Designed a battery-powered PCB integrating **NFC**, **accelerometer**, **LED control**, and a **custom 5-turn, 2.76 µH antenna** for **13.56 MHz NFC**, with **antenna tuning**, **SWD debugging**, and **battery monitoring**.

EDUCATION

University of Massachusetts Lowell **Lowell, MA**
Bachelor of Science in Computer Engineering Expected May, 2028

- Electronics 1, Microprocessors II, Advanced Digital Systems Design, Network Design

North Shore Community College **Lynn, MA**
Associate of Arts in Computer Science August, 2025

- Dean's List Spring 2024

LEADERSHIP AND INVOLVEMENT

River Hawk Robotics Undergraduate Club **Vice President**
University of Massachusetts Lowell April 2025 - April 2026

- Led a **6-member team** in the overhaul of "Sparky," a student-built robot used in the University Engineering Night sled race.
- Proposed **8 local companies** as prospective sponsors to club leadership to support fundraising efforts.